FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Main)-2024

TEST DATE: 29-03-2024

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. B

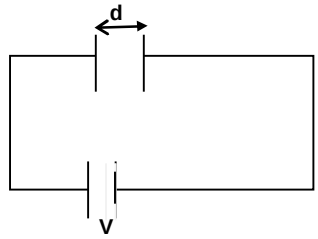
Sol. $C = \dfrac{k\varepsilon_0 A}{d}$

$Q = CV$

$F = \dfrac{q^2}{2\varepsilon_0 A}$

$U = \dfrac{1}{2}CV^2$

$E = \dfrac{V}{d}$



A = Area of each plate

2. C

Sol. Theoretical

3. A

Sol. $\gamma = \alpha_1 + \alpha_2 + \alpha_3$

$= 40 \times 10^{-6} + 3 \times 10^{-6} + 7 \times 10^{-6}$

$= 50 \times 10^{-6}$.

4. C

Sol. Large retentivity and large coercivity is required to make permanent magnet.

5. A

Sol. Light ray will come and from Face we AC if

$\mu \sin i = \mu \sin 60^0 < 1$

Which is possible only for $\mu = 1.1$ so red colour will come out from face AC.

6. C

Sol. Position where $\vec{B}_1$ & $\vec{B}_2$ is calculated are at opposite position from current carrying element.

7. C

Sol. $T_1 = 2\pi\sqrt{\frac{M}{k_1}}$, $k_1 = \frac{4\pi^2 M}{T_1^2}$

$T_2 = 2\pi\sqrt{\frac{2M}{k_1 + 2k_2}}$, $k_1 + 2k_2 = \frac{4\pi^2[2M]}{T_2^2}$

$k_2 = 4\pi^2 M\left[\frac{2}{T_2^2} - \frac{1}{T_1^2}\right]\times\frac{1}{2}$

$T' = 2\pi\sqrt{2M\left(\frac{1}{2k_1} + \frac{1}{k_2}\right)}$

$T' = 2\pi\sqrt{2M\left(\frac{T_1^2}{4\pi^2 M}\frac{1}{2} + \frac{1\times 2}{4\pi^2 M}\left[\frac{T_1^2 T_2^2}{2T_1^2 - T_2^2}\right]\right)}$

$= \sqrt{T_1^2 + \left[\frac{4T_1^2T_2^2}{2T_1^2 - T_2^2}\right]}$.

8. D

Sol. $\Delta\rho = \rho\sqrt{g^2 + 3g^2}\,2R$

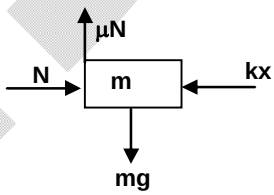
$= 4\,\rho gR$.

9. A

Sol. $2k\,(0.2) = N$, $\mu N = mg$

$\mu 2k\,(0.2) = mg = 200$

$k = \frac{200}{0.2\times 0.2\times 2} = 2500$ N/m



10. B

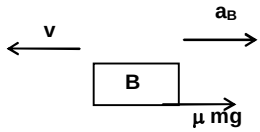
Sol. $a_B = \frac{\mu mg}{m} = \mu g$

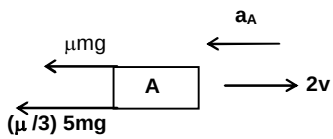
$\mu mg + (\mu/3)\,5mg = 4\,ma_A$

$(8/3)\mu mg = 4\,ma_A$

$a_A = (2/3)\mu g$

$a_A/a_B = 2/3$





11. C

Sol. $[\alpha x] = [F]$

$[\alpha] = \frac{MLT^{-2}}{L} = MT^{-2}$

$$\left[\frac{\alpha}{\sqrt{\beta}}k^2\right] = \{F\}$$

$$\frac{\alpha^2k^4}{\beta} = F^2$$

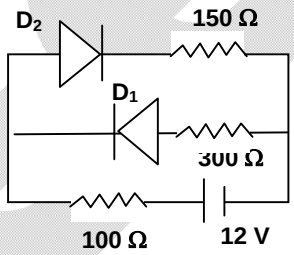
$$\beta = \frac{\alpha^2k^4}{F^2}$$

$$\beta = \frac{[M^2T^{-4}][M^4L^8T^{-8}]}{[M^2L^2T^{-4}]}$$

$$\beta = [M^4L^6T^{-8}]$$

12. D

Sol. Diode D_1 is in reversed biased so current through it will not flow in this branch.



Equivalent resistance = 150 + 100 = 250 Ω

$$I = \frac{12}{250} = \frac{48}{1000} = 48mA$$

13. B

Sol. $u = \sqrt{2gH}$

$v = \sqrt{2gh}$

$v = eu$

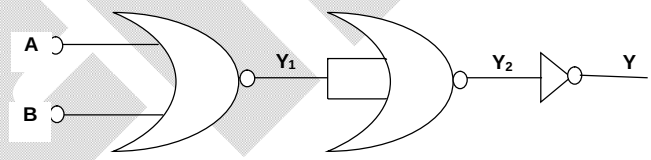
$h = e^2H$ (after first collision)

After 2nd collision = $\left[\left(\frac{1}{2}\right)^{1/8}\right]^4 \times 128 = \frac{128}{\sqrt{2}} = 64\sqrt{2}m$

After 4th collision $\left[\left(\frac{1}{2}\right)^{1/8}\right]^8 \times 128 = \frac{128}{2} = 64m$.

14. C

Sol.



A	B	Y_1	Y_1	Y_2	Y
0	0	1	1	0	1
1	0	0	0	1	0
0	1	0	0	1	0
1	1	0	0	1	0

15. A

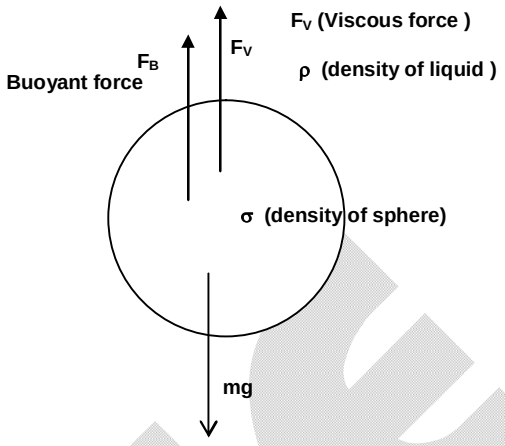
Sol. $F_v = mg - F_B$

$= mg - \rho\left(\frac{m}{\sigma}\right)g$

$= mg\left(1-\frac{\rho}{\sigma}\right)$

$= mg\left(1-\frac{2}{8}\right)$

$= \frac{9}{4}\times 10^{-3}$ N.



16. D

Sol. Rate of heat flow will be maximum if thermal resistance is less

(A) $R_T = \frac{L}{(2k)(R)^2\pi} = \frac{1}{2\pi}\frac{L}{kR^2}$

(B) $R_T = \frac{8L}{(4k)(3R)^2\pi} = \frac{2}{9}\frac{L}{\pi kR^2}$

(C) $R_T = \frac{L/3}{(2k)(R)^2\pi} = \frac{1}{6}\frac{L}{\pi kR^2}$

(D) $R_T = \frac{L}{24\pi kR^2}$

17. C

Sol. Slope $= \frac{dP}{dv}$

PV^{γ} = constant.

$\frac{dP}{dv} = -\gamma\frac{P}{v}$

γ for helium is more.

18. B

Sol. At high frequency capacitor has zero resistance and inductor has infinite resistance. So effective resistance is 5Ω.

I = 220/5 = 44A

19. D

Sol. $\frac{\left(\frac{dU}{dt}\right)_A}{\left(\frac{dU}{dt}\right)_B} = \frac{\sigma 4\pi r_A^2 T_A^4}{\sigma 4\pi r_B^2 T_B^4}$

$10^6 = (4000)^2\,(T_A/T_B)^4$

$T_A/T_B = \frac{1}{2} = \frac{\lambda_B}{\lambda_A}$

$\frac{\lambda_A}{\lambda_B} = 2$.

20. D

Sol. L. C = pitch /No. of division on circular scale.

$\frac{2mm}{200} = 0.01mm$

Diameter of wire = 1 + 67 × 0.01

= 1.67 mm.

Curved surface area = πDL

= 3.14 × $\frac{1.67}{10} \times 6.8 = 3.6$

SECTION – B

21. 4

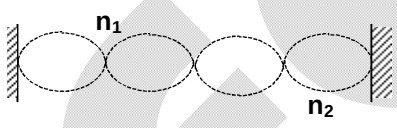
Sol. $n_1(\lambda/2) + (\lambda/4) = 3L/8$

$n_2(\lambda/2) + (\lambda/4) = 5L/8$

divide above equator,

$\frac{2n_1+1}{2n_2+1} = \frac{3}{5}$

$\Rightarrow$ $n_1 = 1, n_2 = 2$



22. 5

Sol. No of deuterium atom is $\frac{2000}{2} \times 6.023 \times 10^{23} = 6.023 \times 10^{26}$

No of reaction $\frac{6.023}{2} \times 10^{26} = 3.011 \times 10^{26}$

Total energy (W) released = $3.01 \times 10^{26} \times 3.2 \times 10^{6} \times 1.6 \times 10^{-19}$

W = pt

t = 5×10^4 years

So x = 5

23. 2

Sol. $T \propto n^3$, $T_1/T_2 = 8 = \frac{64}{8} = \frac{(n_1+1)^3}{(n_2+1)^3}$

$n_1 = 3$, $n_2 = 1$

So, $n_1 - n_2 = 2$.

24. 4

Sol. $a = \frac{v^2-u^2}{2x} = \frac{32-16}{4\times 2} = 2$

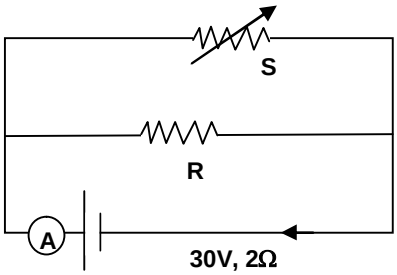
v = u + at = 4 + 2 × 0.5 = 5

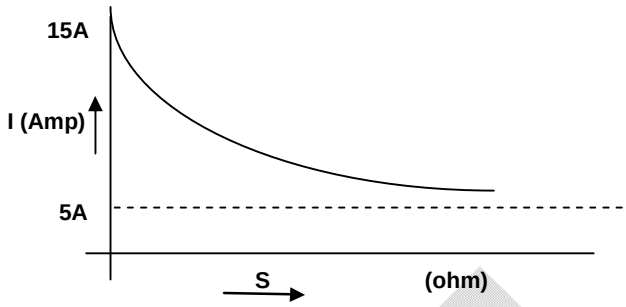
$\frac{dv}{dx} = \frac{a}{v} = \frac{2}{5} = \frac{4}{10}$.

25. 4

Sol. $V - Ir = \frac{IRS}{R+S}$

If S = 0, I = V/r





$V - Ir = \dfrac{IR}{\dfrac{R}{S}+1}$

$S \Rightarrow \infty$

$I = \dfrac{V}{r+R}$

$I = \dfrac{V}{r+\dfrac{RS}{R+S}}$

If $S = 0$, $I = \dfrac{V}{r}$ and $r = 2\Omega$

26. 8

Sol. $i = \alpha t^2 + c$

$t = 0, i = 20$, so $c = 20$

$t = 2, I = 0, 0 = \propto (2)^2 + 20$

so, $\alpha = -5$

flow of charge $= \int_0^2 idt = \left[\dfrac{\alpha t^3}{3} + ct\right]_0^2 = \dfrac{80}{3}C$.

Flow of charge $= \dfrac{\Delta\phi}{R}$

Change in flux $(\Delta\phi) = \dfrac{80}{3}0.3 = 8$

27. 8

Sol. No. of photon incident (X) = power/(hc/λ)

No. of photoelectron emitted (v)= current /e

$$\frac{v}{x} = \frac{\dfrac{\text{current}}{e}}{\dfrac{\text{power}}{hc/\lambda}} = \frac{\text{current}}{\text{power}} \times \frac{hc}{\lambda e}$$

$$= \frac{1}{3.1}\frac{10^{-6} \times 12.4 \times 10^{-7}}{2 \times 10^{-3} \times 2500 \times 10^{-10}}$$

$$\frac{4}{5000} = 8 \times 10^{-4}$$

So, p = 8.

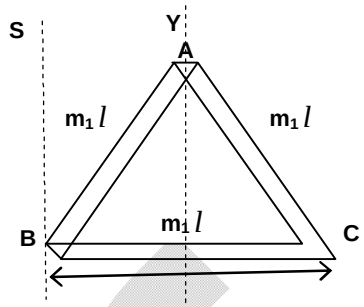
28. 3

Sol. $I_y = \frac{ml^2}{12} + \frac{ml^2}{3} \times 2[\sin^2 30^0]$

$= \frac{ml^2}{12} + \frac{ml^2}{3} \times 2 \times \frac{1}{4} = \frac{ml^2}{12} + \frac{ml^2}{6} = \frac{ml^2}{4}$

$= I_{BS} = I_y + 3m(l/2)^2$

$= \frac{ml^2}{4} + \frac{3ml^2}{4} = ml^2$



29. 9

Sol. $\Delta Q = \Delta U + W$

$Q = \Delta U + Q/6$

$5Q/6 = \Delta U$

$(5/6)nC\Delta T = nc_v\Delta T$

$C = \frac{6}{5}C_v = \frac{6}{5}\frac{R}{\left[\frac{5}{3}-1\right]} = \frac{9R}{5}$.

$A = 9$

30. 3

Sol. $\Delta x = (\mu - 1)t = 0.5\ t$

$I = 2I_0\ (1 + \cos\phi)$

$I = 2I_0 = 2I_0\ (1 - \cos\phi)$

$\cos\phi = 0,\ \phi = \pi/2,$

$\frac{\Delta x}{\lambda} = \frac{\phi}{2\pi}$

$\frac{0.5t}{\lambda} = \frac{\pi/2}{2\pi}$

$t = \lambda/2$

$t = 3000$ Å

Chemistry

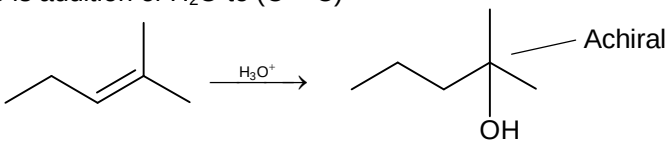
PART – B

SECTION – A

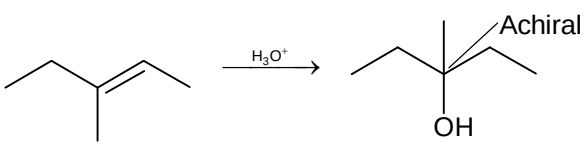
31. D

Sol. There is addition of H_2O to (C = C)

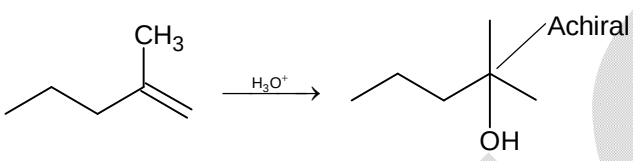
(I)



(II)



(III)

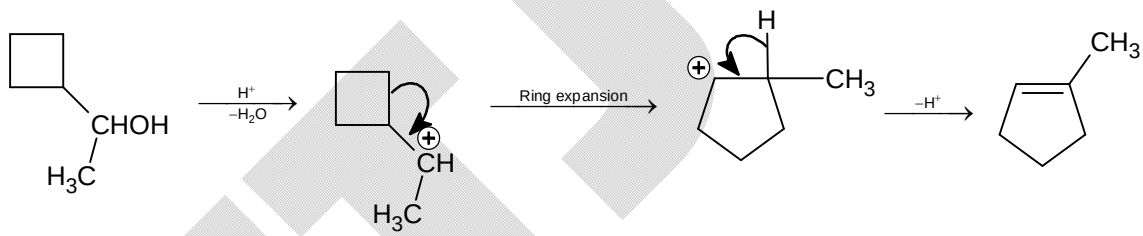


Thus, no optical isomer is formed.

CH_2CH_3

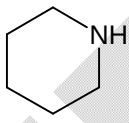
32. C

Sol.



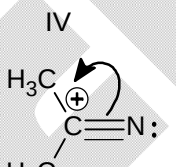
33. D

Sol.

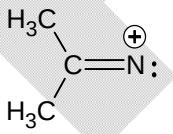


2° amine, maximum basic, lone pair remains intact.

IV



Resonance stabilised



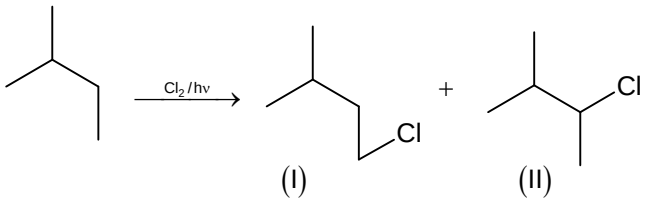
Positive charge on the more electronegative atom, thus unstable.
Thus, it is least basic

34. B

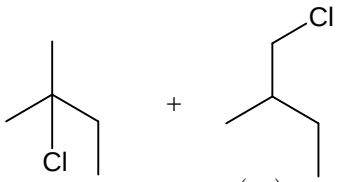
Sol. I is most acidic due to presence of carbonyl grousp. IV is least acid but most basic.
Thus, acidic nature. IV < II < III < I
Increasing pK_a I < III < II < IV.

35. B

Sol.



(I) (II)



(III) (IV)

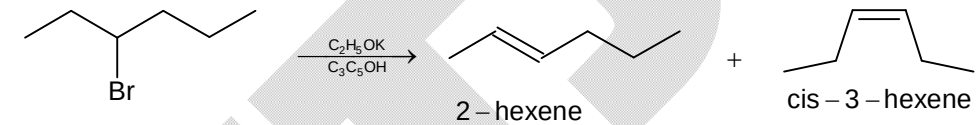
Thus, $C_5H_{11}Cl = 6$

$\therefore N = 6$

On fractional distillation all the isomers are collected as they have different boiling point.
Thus, M = 4.

36. C

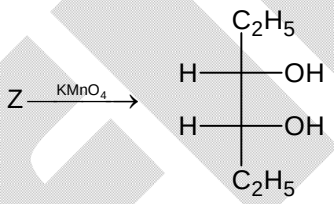
Sol. If meso-diol is formed, it means alkenes must be cis-symmetrical-3-hexene.



3 – bromo – hexene (X)

2 – hexene (Y) + cis – 3 – hexene (Z)

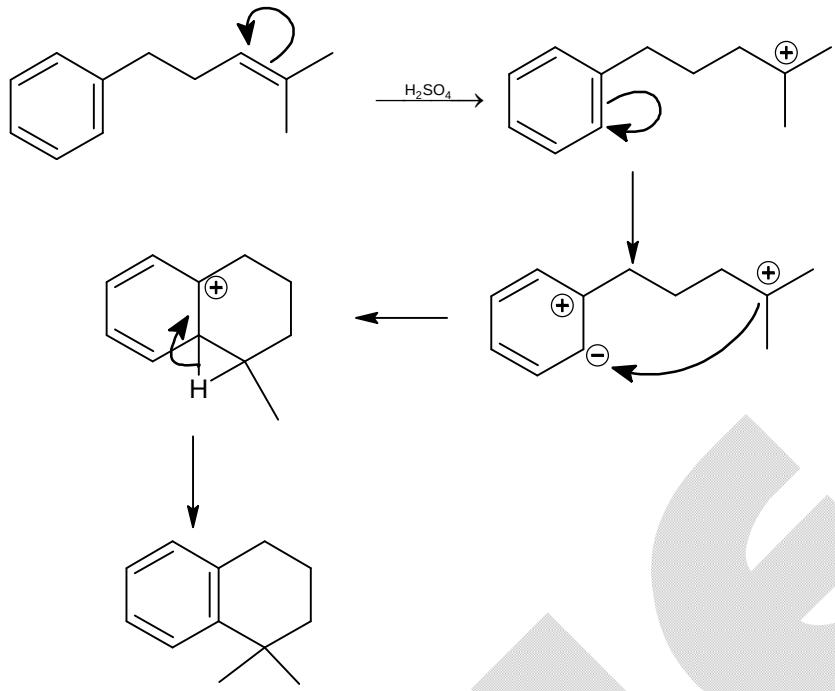
Y and Z are position isomer.



meso by syn – addition

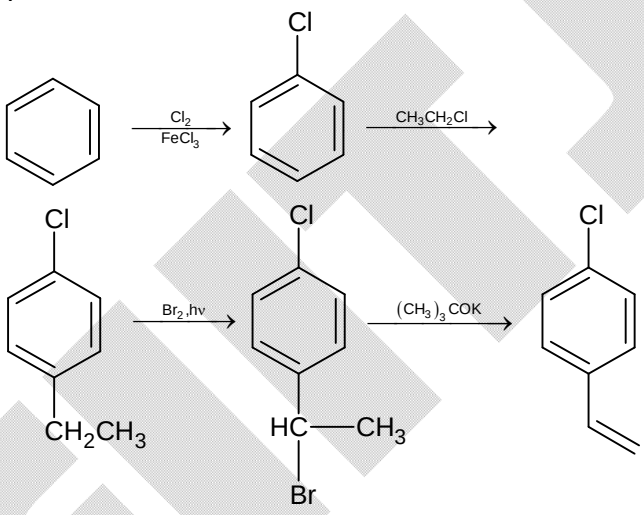
37. C

Sol.



38. A

Sol.



39. B

Sol. $\Delta T_f = i \times K_f \times \text{molality}$

$0.00732 = i \times 1.86 \times 0.002$

$\therefore i = 1.96 = 2$

$\therefore \left[Co(NH_3)_5.(NO_2)\right]Cl \longrightarrow \left[Co(NH_3)_5 NO_2\right]^+ + Cl^-$

40. A

Sol. $E_{Cu/Cu^{2+}} = E^o_{Cu/Cu^{2+}} - \dfrac{0.059}{2}\log\left[Cu^{2+}\right]$

If $\log\left[Cu^{2+}\right] = 0$, i.e. $\left[Cu^{2+}\right] = 1$, then $E_{Cu/Cu^{2+}} = E^o_{Cu/Cu^{2+}}$

or $OA = E^{o}_{Cu/Cu^{2+}} = -E^{o}_{Cu/Cu^{2+}} = -0.34$

Now, $E_{Cu/Cu^{2+}} = -0.34 - \frac{0.059}{2}\log 0.1$

$= -0.34 + \frac{0.059}{2}\text{V}$

41. C

Sol. $E = E^{o} - \frac{0.059}{2}\log\left[H^{+}\right]^{2} = 1.30 - \frac{0.059}{2}\log\left(10^{-2}\right)^{2}$

$= 1.30 + \frac{0.236}{2} = 1.418\ \text{V}$

42. A

Sol.

$$C_4H_8 \xrightarrow{\Delta} 2C_2H_4$$

Mole at $t = 0$	a	0
Mole at $t = t$	$(a - x)$	2x

When $\frac{2x}{a-x} = 1$, then $x = \frac{a}{3}$

$t = \frac{2.303}{2.48 \times 10^{-4}}\log\frac{a}{a - \frac{a}{3}}$

= 1635.2 second = 27.25 minute

43. C

Sol. $2.303\log\frac{K_2}{K_1} = \frac{E_a}{R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$;

$\therefore 2.303\log\frac{K_2}{K_1} = \frac{65 \times 10^3}{8.314}\left[\frac{25}{298 \times 273}\right]$

$\therefore \frac{K_2}{K_1} = 10.4$

44. B

Sol. $4Zn + 10HNO_3\,(\text{very dil.}) \longrightarrow 4Zn(NO_3)_2 + NH_4NO_3 + 3H_2O$

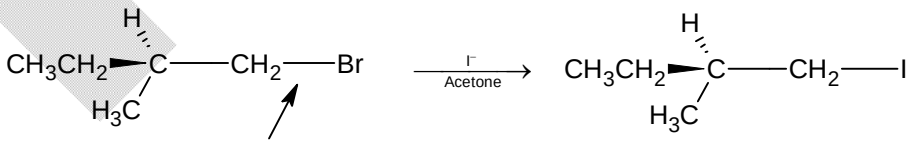
45. A

Sol. $-Br$ and $-C(CH_3)_3$ are in trans position, thus elimination is favoured to maximum extent.

Also dissociation energy (C – Br < C – Cl)

46. C

Sol.



Leaving group is not at chiral carbon.

I^- is a strong nucelophile and is taken in polar aprotic solvent. Thus S_N2 reaction. As there is no attack on chiral centre, configuration of the reactant and product is same.

47. A

Sol. $\bar{\nu} = \dfrac{1}{\lambda} = \bar{R}_H Z^2 \left[\dfrac{1}{n_1^2} - \dfrac{1}{n_2^2}\right]$

For a given series

$\lambda \propto \dfrac{1}{Z^2}$

$\therefore \dfrac{\lambda(H)}{\lambda(Li^{2+})} = \left[\dfrac{Z(Li^{2+})}{Z(H)}\right]^2 \Rightarrow \dfrac{821}{\lambda(Li^{2+})} = \left(\dfrac{3}{1}\right)^2 = 9$

$\therefore \lambda(Li^{2+}) = \dfrac{821}{9} = 91.2 \text{ nm}$

48. C

Sol.

	$N_2(g)$	$+ 3H_2(g) \rightleftharpoons$	$2NH_3(g)$
Initial	1	3	0
At equil.	$(1-x)$	$3(1-x)$	2x

$2NH_3 + H_2SO_4 \rightleftharpoons (NH_4)_2 SO_4$

Moles of H_2SO_4 required $= \dfrac{MV}{1000} = \dfrac{250 \times 1}{1000} = 0.250$

$\therefore$ Molarity $(\text{mol L}^{-1}) \times \text{mL} = \text{millimoles}$

Since, $1H_2SO_4 \equiv 2NH_3$

Thus, $NH_3 = 0.500$ mol

$\therefore 2x = 0.500 \Rightarrow x = 0.250$

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3} = \frac{\left(\frac{2x}{1}\right)^2}{\frac{(1-x)}{1}\left(\frac{1-x}{1}\right)^3 (3)^3}$$

$$= \frac{(0.50)^2}{(0.750)^4 (27)} = 0.029$$

49. D

Sol. (A) Thus, on decreasing volume, pressure is increased.
Thus, formation of product is increased (and thus deceasing CO)

(B) Addition of argon (inert gas) increases pressure thus as in (A).

(C) O_2 (reactant) will react CO hence amount of CO decreases.

50. B

Sol. Under adiabatic expansion, temperature falls and internal energy decreases.

$q + W = \Delta E, q = 0$

$\therefore \ -\Delta E = W$

Decrease, $-W = nC_V (T_2 - T_1)$

$-3 \times 10^3 \text{ J} = 1 \text{ mol} \times 20 \text{ J K}^{-1} \text{ mol}^{-1} (T_2 - 300 \text{ K})$

$T_2 = 150 \text{ K}$

SECTION – B

51. 2

Sol. $\Delta T_f(KCN) = K_f mi$

$0.744 = K_f(0.2) \times 2$

$\therefore K_f = 1.86^o \text{ molal}^{-1}$

On adding $Hg(CN)_2$,

$2KCN + Hg(CN)_2 \rightleftharpoons K_2[Hg(CN)_4]$

0.2 mol 0.1 mol 0.1 mol

Thus, $K_2[Hg(CN)_4]$ is 0.1 molal

$\therefore \Delta T_f(\text{complex}) = (+0.744 - 0.372)^o = 0.372^o$

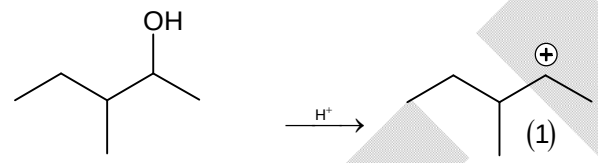
$\Delta T_f(\text{complex}) = K_f(0.1) \times i$

$\therefore 0.372 = 1.86(0.1) \times i$

$\Rightarrow i = \dfrac{0.372}{1.86 \times 0.1} = 2$

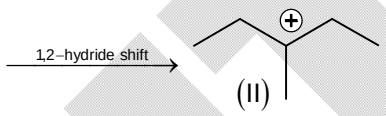
52. 5

Sol.

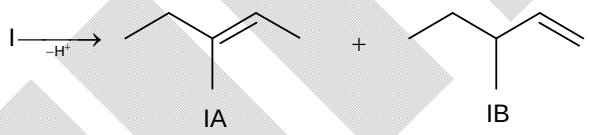


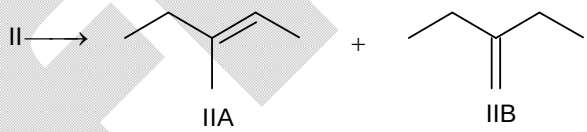
1,2 – hydride shift

2° carbocation

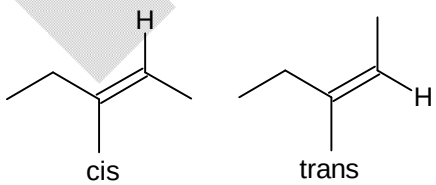


3° carbocation

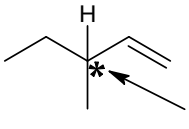




IA can exist as cis and trans.



IB can exist as d(+) and $\ell(-)$



IA and IIA are same. Total number of isomers = 5.
IA (C_1, T) IB ((d, ℓ) and IIB

53. 78

Sol.

$$CH \equiv C(CH_2)_n - C \equiv CH$$
$$(A)$$

$$\downarrow BH_3 / H_2O_2, OH^-$$
(addition of H_2O followed by tautomerisation)

$$OHC - CH_2 - (CH_2)_n - CH_2 - CHO$$
$$(B)$$

Molar mass of A = (14n + 50)
Molar mass of B = (14n + 86)
1.14 g of B is from = 0.78 g of A

$$\therefore (14n+86)\text{g of B is from } = \frac{0.78\times(14n+80)}{(1.14)}\text{g of A}$$

(Molar mass of A)

$$\therefore \frac{0.78}{1.14}(14n+86) = (14n+50)$$

$$\therefore n = 2$$

Thus, alkyne is

$$CH \equiv C - (CH_2)_2 - C \equiv CH$$

$$\text{Molar mass } = 14n + 50 = 78\,\text{g mol}^{-1}$$

54. 22

Sol.

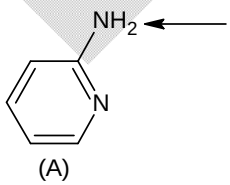
$$\mu_{CH_3-Cl}(\text{theoretical}) = e \times l$$
$$= 4.8\times10^{-10}\times1.78\times10^{-8}\ \text{esu}-\text{cm}$$
= 8.544 Debye (D)

$$\%\text{ionic character} = \frac{\text{Experimental }\mu}{\text{Theoretical }\mu}\times100$$

$$= \frac{1.87}{8.544}\times100$$
= 22%

55. 10

Sol.



Only this N can be estimated by the given method.

(A)

$$\text{Molar mass of A}(C_5H_6N_2) = 94.0\ \text{g/mol}^{-1}$$

$\frac{0.94}{94} = 0.01\,\text{mol}$

= 0.01 equivalent

$0.5\,\text{M}\ H_2SO_4 = 1.0\ \text{N}\ H_2SO_4$

$\frac{NV}{1000} = 0.01$

$\frac{1\times V}{1000} = 0.01$

V = 10 mL

56. 1

Sol. $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 \downarrow + H_2O$

$n(CO_2) = \frac{PV}{RT}$

$= \frac{0.1\,\text{bar} \times 0.1\,\text{L}}{0.083\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times 302\ \text{K}}$

$= 4\times10^{-4}$ mol

$Ca(OH)_2 = 200$ mL of $0.01\,\text{M} = 2$ millimol $= 2\times10^{-3}$ mol

$Ca(OH)_2$ reacted $= 4\times10^{-4}$ mol

$Ca(OH)_2$ unreacted $= 2\times10^{-4} - 4\times10^{-4} = 16\times10^{-4}$ mol

$\left[OH^-\right] = 2\times16\times10^{-4}$ mol in 200 mL

(CO_2 is in gaseous state and its volume is not added)

$\therefore \left[OH^-\right] = \frac{32\times10^{-4}\ \text{mol}}{0.2\ \text{L}} = 0.016\ \text{M}$

$pOH = -\log(0.016) = 1.8$

$\therefore pH = 14.0 - 1.8 = 12.2 = 1.22\times10^1$

57. 2

Sol. Two HCHO molecule + Three HCOOH + One CO_2 is formed.

58. 15

Sol. $NH_2CONH_2 \xrightarrow{NaOBr} N_2$, 22400 mL at NTP = 60 g urea

$5.6\times10^{-3}\ \text{mL} = \frac{60}{22400}\times5.6\times10^{-3}\ \text{g/mL}$ of body fluid

= 15 g urea ppm,

59. 4

Sol. β – Keto acid decarboxylate on heating.

60. 4

Sol. Magnetic moment $= \sqrt{n(n+2)} = 4.89$ BM

$= \sqrt{4\times6} = 4.89$ BM

Mathematics

PART – C

SECTION – A

61. C

Sol. $x^2 - 2x\cos\theta + 1, = 0$

$\therefore x = \dfrac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2} = \cos\theta \pm i\sin\theta$

Let $x = \cos\theta + i\sin\theta$

$\therefore x^{2n} - 2x^n \cos n\theta + 1$

$= \cos 2n\theta + i\sin 2n\theta - 2(\cos n\theta + i\sin n\theta)\cos n\theta + 1$

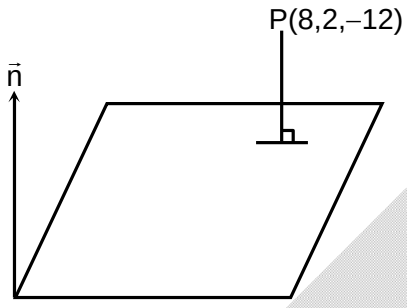
$= \cos 2n\theta + 1 - 2\cos^2 n\theta + i(\sin 2n\theta - 2\sin n\theta \cos n\theta)$

$= 0 + i\,0 = 0$

62. B

Sol. $\vec{n} = 7\hat{i} + 2\hat{j} - \hat{k}$ is normal to plane

(Assuming $n = a\hat{i} + b\hat{j} + c\hat{k}$ and using $\vec{n}.\overrightarrow{AB} = 0, \vec{n}.\overrightarrow{BC} = 0, \vec{n}.\overrightarrow{AC} = 0$)



P = (8, 2, –12)

$\overrightarrow{AP} = 6\hat{i} + 3\hat{j} - 16\hat{k}$

$\therefore$ Distance $d = \left|\dfrac{\overrightarrow{AP}.\vec{n}}{|\vec{n}|}\right| = \left|\dfrac{42 + 6 + 16}{\sqrt{49 + 4 + 1}}\right| = \dfrac{64}{\sqrt{54}} = \dfrac{64}{3\sqrt{6}} = \dfrac{64\sqrt{6}}{18} = \dfrac{32\sqrt{6}}{9}$

63. A

Sol. For real roots $D \geq 0$

$(k - 2)^2 - 4(k^2 + 3k + 5) \geq 0$

$(k^2 + 4 - 4k) - 4k^2 - 12k - 20 \geq 0$

$-3k^2 - 16k - 16 \geq 0$; $3k^2 + 16k + 16 \leq 0$

$\left(k + \dfrac{4}{3}\right)(k + 4) \leq 0$

Now $E = \alpha^2 + \beta^2$; $E = (\alpha + \beta)^2 - 2\alpha\beta$

$E = (k - 2)^2 - 2(k^2 + 3k + 5) = -k^2 - 10k - 6$

$E = -(k^2 + 10k + 6) = -[(k + 5)^2 - 19] = 19 - (k + 5)^2$

$E_{min} = 50/9$ at $k = -4/3$

$E_{max} = 18$ at $k = -4$

64. A

Sol. $\dfrac{20!}{p!q!r!}(2x)^n(-y)^q(z)^r = \dfrac{20!}{p!q!r!}2^p(-1)^q x^p y^q z^r$

$p + q + r = 20$, $q = 0$

p + r = 20 (p is even and r is odd)
even + odd = even (never possible)
Coefficient of such power never occur
$\therefore$ coefficient is zero

65. A

Sol. Starting with $1\boxed{1}\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\ 23456789 = {}^8C_4 = 70$

Starting with $2\boxed{2}\boxed{\ }\boxed{\ }\boxed{\ }\boxed{\ }\ 3456789 = {}^7C_4 = 35$

Total = 105
$(105)^{th}$ number 26789

66. B

Sol. n = 3, P (success) = P(HT or TH) = $\frac{1}{2}$

$\Rightarrow$ p = q = $\frac{1}{2}$ and r = 2

P(r = 2) = ${}^3C_2\left(\frac{1}{2}\right)^2 \cdot \frac{1}{2} = \frac{3}{8}$

67. A

Sol. $(\vec{a}.\vec{c})\vec{b} - (\vec{a}.\vec{b})\vec{c} = \frac{1}{\sqrt{2}}\vec{b} + \frac{1}{\sqrt{2}}\vec{c}$

$\therefore \quad \vec{a}.\vec{c} = \frac{1}{\sqrt{2}}$ and $\vec{a}.\vec{b} = -\frac{1}{\sqrt{2}}$

$\Rightarrow$ angle between $\vec{a}$ and $\vec{c} = \frac{\pi}{4}$ and angle between $\vec{a}$ and $\vec{b} = \frac{3\pi}{4}$

68. D

Sol. $\frac{\int_0^1 (1-x^p)^q}{\int_0^1 (1-x^p)^{q+1}} = \frac{p(q+1)+1}{p(q+1)}$

69. C

Sol. The coordinates of points P, Q, R are (–1, 0), (0, 0), $(3, 3\sqrt{3})$ respectively.

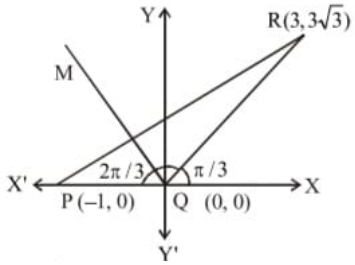
Slope of QR = $\sqrt{3}$

$\Rightarrow \quad \tan\theta = \sqrt{3}$

$\Rightarrow \quad \theta = \frac{\pi}{3}$

$\Rightarrow \quad \angle RQX = \frac{\pi}{3}$

$\therefore \quad \angle RQX = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$



Let QM bisects the $\angle PQR$

$\therefore$ Slope of the line QM = $\tan\frac{2\pi}{3} = -\sqrt{3}$

$\therefore$ Equation of line QM is (y – 0) = $-\sqrt{3}$ (x – 0)

$\Rightarrow$ $y = -\sqrt{3}x \Rightarrow \sqrt{3}x + y = 0$

70. A

Sol. $\lim\limits_{n\to\infty}\prod\limits_{r=3}^{n}\dfrac{(r-2)}{(r+2)}\times\dfrac{(r^2+4+2r)}{(r^2+4-2r)}$

$\Rightarrow \lim\limits_{n\to\infty}\left\{\dfrac{1}{5}\times\dfrac{2}{6}\times\dfrac{3}{7}\times\dfrac{4}{5}\times\dfrac{5}{4}\times\dfrac{6}{10}.....\right\}\times\left\{\dfrac{19\times28\times.....}{7\times12\times19\times28}\right\}=\dfrac{1\times2\times3\times4}{7\times12}=\dfrac{2}{7}$

71. A

Sol. $t_k=\cot^{-1}\left[\dfrac{1+k(k+1)^2(k+2)}{2(k+1)}\right]=\tan^{-1}\left[\dfrac{2(k+1)}{1+k(k+1)^2(k+2)}\right]=\tan^{-1}\left[\dfrac{(k+1)(k+2)-(k)(k+1)}{1+k(k+1)^2(k+2)}\right]$

$t_k=\tan^{-1}((k+1)(k+2))-\tan^{-1}(k(k+1))$

$\Rightarrow \sum\limits_{k=1}^{n}t_k=\tan^{-1}((n+1)(n+2))-\tan^{-1}(1(1+1))=\tan^{-1}((n+1)(n+2))-\tan^{-1}(2)$

72. A

Sol. $S=\sum\limits_{K=1}^{2009}f(K)=f(1)+f(2)\ldots\ldots f(2009)$

Since $f(1)=f(2010)$

$S=f(2)+f(3)\ldots\ldots f(2010)$

$=\sum\limits_{K=1}^{2009}f(K+1)=\sum\limits_{K=1}^{2009}\left((-1)^{K+1}K-2f(K)\right)=\sum\limits_{K=1}^{2009}(-1)^{K+1}K-2\sum\limits_{K=1}^{2009}f(K)$

$\Rightarrow S=[2009-2008+2007-2006\ldots..1-0]-2S$

$\Rightarrow 3S=1005\Rightarrow S=335$

73. A

Sol. For an onto function to exist from $A_r\to B_r$

$|A_r|\geq|B_r|\geq1$

No of ways $=\sum\limits_{i=1}^{m}{}^{m}C_i\sum\limits_{j=i}^{m}{}^{m}C_j$

We can select B_r of size i in ${}^{n}C_i$ ways and A_r can then have any size from {i, i + 1, …. m} which can be selected in

$\sum\limits_{j=i}^{m}{}^{m}C_j$ ways

$\sum\limits_{i=1}^{m}{}^{m}C_i\sum\limits_{j=i}^{m}{}^{m}C_j=\sum\limits_{i=0}^{m}{}^{m}C_i\sum\limits_{j=i}^{m}{}^{m}C_j-{}^{m}C_0\left[{}^{m}C_0+{}^{m}C_1.....{}^{m}C_m\right]$

$=\left[\left({}^{m}C_0\right)^2+\left({}^{m}C_1\right)^2.....\left({}^{m}C_m\right)^2\right]+\sum\limits_{r<s}{}^{m}C_r\,{}^{m}C_s-2^m$

$={}^{2m}C_m+\dfrac{1}{2}\left[2^{2m}-{}^{2m}C_m\right]-2^m=\dfrac{1}{2}\left[2^{2m}-2^{m+1}+{}^{2m}C_m\right]$

74. B

Sol. $\int\dfrac{\left(2+\sqrt{x}\right)dx}{\left(x+1+\sqrt{x}\right)^2}=\int\dfrac{\dfrac{2}{x^2}+\dfrac{1}{x^{3/2}}}{\left(\dfrac{1}{x}+\dfrac{1}{\sqrt{x}}+1\right)^2}dx$

Let $\dfrac{1}{x}+\dfrac{1}{\sqrt{x}}+1=t\Rightarrow\left(-\dfrac{1}{x^2}-\dfrac{1}{2x^{3/2}}\right)dx=dt$

75. B

Sol. Replacing $\cos^2\frac{x}{4}-\sin^2\frac{x}{4}$ by

$\cos\frac{x}{2}$; we get $\frac{dy}{dx}=\frac{1}{2}(k^2-3k+2)\sin\frac{x}{2}+(k-1)$

$=-\frac{1}{2}(k-1)(k-2)\sin\frac{x}{2}+(k-1)=-\frac{(k-1)}{2}\left[(k-2)\sin\frac{x}{2}-2\right]\neq 0$

Because the function does not posses critical point $k-1\neq 0$ and $\sin\frac{x}{2}=\frac{2}{k-2}$ does not posses any solution.

$\Rightarrow$ $k\neq$ and $\left|\frac{2}{k-2}\right|>1$ as $\left|\sin\frac{x}{2}\right|\leq 1$ $\Rightarrow k\neq$ and $|k-2|<2$

$\Rightarrow$ $k\neq 1$ and $-2<(k-2)<2$ or $0<k<4$

Thus k lies in the interval (0, 4) but $k\neq 1$ i.e. $k\in(0,1)\cup(1,4)$

76. B

Sol. The locus is the pair of planes passing through the point of intersection of the lines and having their bisectors as the normal.

77. A

Sol. $x(2x^2y^3+1)dy=y(1-3x^2y^3)dx$

$\Rightarrow$ $x^2y(2xdy+3ydx)=\frac{ydx-xdy}{y^2}$

$\Rightarrow$ $d\left(x^3y^2\right)=d\left(\frac{x}{y}\right)$ $\Rightarrow x^3y^2=\frac{x}{y}+c$ $\Rightarrow x^3y^3=x+cy$

78. A

Sol. $\lim\limits_{x\to 0}\frac{(\tan x)^{3/2}\left[1-(\cos x)^{3/2}\right]}{x^{3/2}\cdot x^2}$

$=1\times\lim\limits_{x\to 0}\frac{1-\cos^3 x}{x^2}\cdot\frac{1}{1+(\cos x)^{3/2}}$

$=\frac{1}{2}\cdot\frac{1}{2}(1+\cos x+\cos^2 x)=\frac{3}{4}$

79. C

Sol. $a_n=\int\limits_0^{\pi/2}(1-\sin t)^n\sin 2t\,dt$

Let $1-\sin t=u\Rightarrow -\cos t\,dt=du$

$=2\int\limits_0^1 u^n(1-u)\,du=2\left(\int\limits_0^1 u^n du-\int\limits_0^1 u^{n+1}du\right)=2\left(\frac{1}{n+1}-\frac{1}{n+2}\right)$

Hence, $\frac{a_n}{n}=2\left(\frac{1}{n(n+1)}-\frac{1}{n(n+2)}\right)$

$\lim\limits_{n\to\infty}\sum\limits_1^n\frac{a_n}{n}=2\left(\frac{1}{n(n+1)}-\frac{1}{n(n+2)}\right)$

$\lim\limits_{n\to\infty}\sum\limits_1^n\frac{a_n}{n}=2\left(\sum\left(\frac{1}{n}-\frac{1}{n+1}\right)\right)-\frac{1}{2}\sum\left(\frac{1}{n}-\frac{1}{n+2}\right)$

$$= 2\left(\sum_1^n\left(\frac{1}{n}-\frac{1}{n+1}\right)\right)-\sum_1^n\left(\frac{1}{n}-\frac{1}{n+2}\right)$$

$$= 2(1)-\left[\left(1-\frac{1}{3}\right)+\left(\frac{1}{2}-\frac{1}{4}\right)+\left(\frac{1}{3}-\frac{1}{5}\right)+.....\right] = 2-\frac{3}{2}=\frac{1}{2}$$

80. D

Sol. A : blood result says positive about the disease

B_1 : Person suffers from the disease : $P(B_1) = \frac{1}{100}$

B_2 : Person does not suffer from the disease : $P(B_1) = \frac{99}{100}$

$P(A/B_1) = \frac{99}{100}$, $P(A/B_2) = \frac{1}{100}$

$$P(B_1/A) = \frac{P(B_1).P(A/B_1)}{P(B_1).P(A/B_1)+P(B_2).P(A/B_2)}$$

$$= \frac{\frac{1}{100}\cdot\frac{99}{100}}{\frac{1}{100}\cdot\frac{99}{100}+\frac{99}{100}\cdot\frac{1}{100}} = \frac{99}{2(99)} = \frac{1}{2} = 50\%$$

SECTION – B

81. 343

Sol. $\vec{V} = \vec{A}\times\left[(\vec{A}\cdot\vec{B})\vec{A}-(\vec{A}\cdot\vec{A})\vec{B}\right]\cdot\vec{C}$

$$= \left(\underbrace{\vec{A}\times(\vec{A}\cdot\vec{B})\vec{A}}_{\text{zero}}-(\vec{A}\cdot\vec{A})\vec{A}\times\vec{B}\right)\cdot\vec{C} = -|\vec{A}|^2\left[\vec{A}\ \ \vec{B}\ \ \vec{C}\right]$$

Now, $|\vec{A}|^2 = 4 + 9 + 36 = 49$

$$\left[\vec{A}\ \ \vec{B}\ \ \vec{C}\right] = \begin{vmatrix}2 & 3 & 6\\ 1 & 1 & -2\\ 1 & 2 & 1\end{vmatrix} = 7$$

$$\therefore \left|-|\vec{A}|^2\left[\vec{A}\ \ \vec{B}\ \ \vec{C}\right]\right| = 49\times 7 = 343$$

82. 12

Sol. Let the observations be x_1, x_2, x_3, x_4, x_5 and x_6, so their mean $\bar{x} = \frac{\sum_{i=1}^{6} x_i}{6} = 8 \Rightarrow \sum_{i=1}^{6} x_i = 48$

On multiplying each observation by 3, we get the new observations as $3x_1$, $3x_2$, $3x_3$, $3x_4$, $3x_5$ and $3x_6$.

Now, their mean = $\bar{x} = \frac{\sum_{i=1}^{6} 3x_i}{6} = \frac{3\times 48}{6} = 24$

Variance of new observations = $\bar{x} = \frac{\sum_{i=1}^{6}(3x_i-24)^2}{6} = \frac{3^2\sum_{i=1}^{6}(x_i-8)^2}{6}$

$= \frac{9}{1} \times$ Variance of old observations $= 9 \times 4^2 = 144$

Thus, standard deviation of new observations $= \sqrt{\text{variance}} = \sqrt{144} = 12$

83. 1

Sol. $a^2(1 + b^2) \geq 2a^2b$ (using A.M. – G.M.)

$b^2(1 + c^2) \geq 2b^2c$

$c^2(1 + a^2) \geq 2c^2a$

Using A.M. – G.M. again

$$\frac{2a^2b + 2b^2c + 2c^2a}{3} \geq (8a^3b^3c^3)^{1/3}$$

$2a^2b + 2b^2c + 2c^2a \geq 6abc$

and the equality holds $a = b = c = 1$

84. 31

Sol. $z_r = \frac{z_n - 1}{2}$

$z^4 + z^3 + 2 = 0$

$2z_r + 1 = z_n$

$$\frac{(z_n - 1)^4}{16} + \frac{(z_n - 1)^3}{8} + 2 = 0$$

$(z_n - 1)^4 + 2(z_n - 1)^3 + 32 = 0$

Product of roots = 31

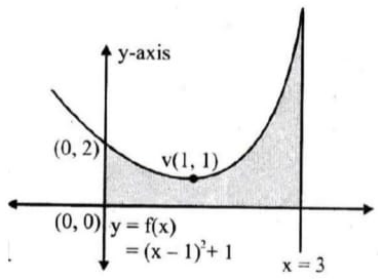
85. 6

Sol. We have $y = f(x)$, so $\frac{dy}{dx} = f'(x)$

Now, $\frac{dy}{dx} + \left(\frac{-1}{x}\right)y = \left(x - \frac{2}{x}\right)$ (Linear differential equation)

$\therefore$ I.F. $= e^{-\int \frac{1}{x}dx} = e^{-\ln x} = \frac{1}{x}$

Now, general solution is $y\left(\frac{1}{x}\right) = \int\left(x - \frac{2}{x}\right)\frac{1}{x}dx$



$\therefore \quad \frac{y}{x} = x + \frac{2}{x} + c$

As $y(1) = 1 \Rightarrow c = -2$

$\therefore \quad \frac{y}{x} = x + \frac{2}{x} - 2 \Rightarrow f(x) = x^2 - 2x + 2 = (x - 1)^2 + 1$

$\therefore$ Required area $= \int_0^3 \left((x-1)^2 + 1\right)dx = \left(\frac{1}{3}(x-1)^3\right)_0^3 + (x)_0^3 = \left(\frac{8}{3} + \frac{1}{3}\right) + 3 = 3 + 3 = 6$

86. 6

Sol. $AB = \begin{bmatrix} 5x & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10x-2 & 5x \end{bmatrix}, (AB)^2 = \begin{bmatrix} 25x^2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 50x^2-2 & 25x^2 \end{bmatrix} = I$

$= 25x^2 = 1 \Rightarrow 5x = -1 \quad (\because AB \neq I)$

$Tr(AB + (AB)^2 + \ldots\ldots + (AB)^6) = 6$

87. 4

Sol. Let $\alpha = e^{iA}$, $\beta = e^{iB}$ and $\gamma = e^{iC}$ then

$\alpha\beta\gamma = e^{i(A+B+C)} = e^{i\pi} = \cos\pi + i\sin\pi = -1$

and $\Delta = \begin{vmatrix} 1/\alpha^2 & \gamma & \beta \\ \gamma & 1/\beta^2 & \alpha \\ \beta & \alpha & 1/\gamma^2 \end{vmatrix} = \frac{1}{\alpha^2\beta^2\gamma^2}\begin{vmatrix} 1 & \alpha^2\gamma & \alpha^2\beta \\ \beta^2\gamma & 1 & \beta^2\alpha \\ \gamma^2\beta & \gamma^2\alpha & 1 \end{vmatrix} = \begin{vmatrix} 1 & -\alpha/\beta & -\alpha/\gamma \\ -\beta/\alpha & 1 & -\beta/\gamma \\ -\gamma/\alpha & -\gamma/\beta & 1 \end{vmatrix} = \frac{1}{\alpha\beta\gamma}\begin{vmatrix} \alpha & -\alpha & -\alpha \\ -\beta & \beta & -\beta \\ -\gamma & -\gamma & \gamma \end{vmatrix}$

[multiplying $C_1\{C_2\}[C_3]$ by $\alpha\{\beta\}[\gamma]$

$\frac{\alpha\beta\gamma}{\alpha\beta\gamma}\begin{vmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 0 & -1 & -1 \\ 0 & 1 & -1 \\ -2 & -1 & 1 \end{vmatrix} = -2(1+1) = -4$

$\Rightarrow |\Delta| = 4$

88. 3

Sol. Equation of plane passing through (1, 1, 1) and perpendicular to the given line is $x + 2y + 3z = 6$.

The minimum distance of the point (1, 1, 1) from the plane $x + y + z = 1$ measured perpendicular to $\frac{x-x_1}{1} = \frac{y-y_1}{2} = \frac{z-z_1}{3}$ is the perpendicular distance of the point (1, 1, 1) from the line of intersection of the above two places.

The equation of line of intersection of these planes is $\frac{x+4}{1} = \frac{y-5}{-2} = \frac{z-0}{1} = r \quad \ldots(1)$

The minimum distance of (1, 1, 1) from the line (1) is $\frac{2}{3}\sqrt{21}$ units.

89. 5

Sol. For all α, $2a^2 + \alpha a + 3\alpha = b$

Let the polynomial $p(\alpha) = (a + 3)\alpha$

Rearranging the above equation gives,

$p(\alpha) = b - 2a^2$, which is a constant

Therefore, $\deg[p(\alpha)] = 0$

So, a must be -3

Now. $p(\alpha) =)(-3 + 3)\alpha = 0 = b - 2a^2$

$\therefore \quad b - 2(-3)^2 = 0 \Rightarrow b = 18$

$\therefore \quad a = -3, b = 18$

$\therefore \quad a + b = -3 + 18 = 15$

$\therefore \quad \frac{a+b}{3} = \frac{15}{3} = 5$

90. 5

Sol. In radius of (ΔOCF) = 1

$\Rightarrow$ Distance of $\frac{x}{6}-\frac{y}{b}=1$ from (1, –1) is 1 $\Rightarrow$ b = 2.5

OF = 6 = ae

and $b^2 = a^2(1-e^2)$ $\Rightarrow b^2 = a^2 - (ae)^2$

$\Rightarrow a=\sqrt{b^2+(ae)^2}$ = 6.5

AB = 2a and CD = 2b

$\therefore$ AB.CD = $13 \times 5 = 65$

$\therefore$ $\frac{AB\cdot CD}{13}=\frac{65}{13}$ = 5

